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# Unlocking Efficiency: Chemical Gas Shutoff Innovations in High GOR Wells: A Pilot Study in Abu Dhabi, UAE

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## Abstract

This paper presents the findings of a pilot study conducted to optimize well and reservoir performance in a relatively low-permeability heterogeneous carbonate reservoir that has been undergoing Water Alternating Gas (WAG) injection since the early phase of production. The reservoir under study is a thin saturated reservoir developed with a number of horizontal wells in inverted 5 spot pattern or line drive mechanism. The primary challenge addressed in this study was the tendency of over gas production of high gas/oil ratios (GOR), which can significantly impair oil recovery and operational efficiency. To mitigate this issue, the study implemented targeted chemical gas shutoff interventions aimed at reducing gas influx from specific reservoir intervals.

The main objectives of the pilot were to enhance oil recovery, stabilize production rates, and extend the productive life of the wells, all while minimizing inefficiencies caused by complex reservoir dynamics. The operational workflow began with a baseline production logging campaign, which was essential for identifying zones with excessive gas contribution. Advanced logging tools, including spinner flowmeters and optical sensors, were used to accurately characterize the flow profile and isolate high-GOR intervals.

Following the diagnostic phase, a chemical treatment was applied to the identified zones. The selection of chemical agents was based on their compatibility with reservoir conditions and their ability to selectively block gas flow without impairing oil production. After the treatment, a second production logging run was conducted to evaluate the effectiveness of the intervention. The results showed a significant reduction in gas contribution from the treated zones, as confirmed by the logging data. Additionally, oil production either stabilized or improved, indicating that the chemical shutoff successfully reduced gas interference and enhanced reservoir drawdown.

The success of this pilot project was attributed to meticulous planning, the use of high-resolution production logging data, and the careful selection of chemical agents. These factors collectively contributed to the effectiveness of the gas shutoff strategy and demonstrated its potential as a viable solution for managing high GOR in WAG-injected carbonate reservoirs.

## Introduction

Maintaining reservoir pressure through gas and water injection is a cornerstone of enhanced oil recovery strategies, particularly in carbonate reservoirs. This approach supports reservoir energy and helps sustain production rates over time. However, wells located near the gas-oil contact (GOC), especially in crestal regions, often face significant challenges related to elevated gas/oil ratios (GOR). These wells typically have completion intervals situated just below the GOC, making them highly susceptible to gas cusping— a phenomenon where gas from the gas cap is drawn downward into the wellbore due to pressure differentials.

Horizontal wells often experience uneven drawdown, especially in heterogeneous reservoirs, along the lateral length of the well due to variation in permeability, pressure distribution, and fluid contacts. This imbalance leads to the so-called heel-to-toe effect, where the heel (near the wellbore entry) tends to dominate inflow because of lower pressure losses, while the toe contributes less. In reservoirs with a nearby gas cap, the heel zone being exposed to stronger pressure gradients can preferentially draw gas, causing premature gas breakthrough and excessive gas production at the expense of oil. Over time, this gas influx not only reduces oil recovery efficiency but also increases GOR, accelerates reservoir energy depletion, and complicates production management. Effective mitigation requires careful well placement, optimized completion design, and sometimes inflow control devices to balance drawdown along the wellbore.

Gas cusping is primarily driven by the proximity of the well to the GOC and the magnitude of drawdown pressure during production. When drawdown is sufficient, it disrupts the gravity equilibrium and allows free gas to migrate into the oil-producing zone. These results in increased gas production, reduced oil mobility, and ultimately, a decline in oil recovery efficiency. The problem is exacerbated by the high mobility and low viscosity of gas compared to oil, which enables gas to move more readily through the reservoir matrix.

Managing gas production in such wells requires a careful balance between maintaining production targets and minimizing gas interference. Once gas cusping occurs, it is difficult to reverse, and the well may become dominated by gas flow, leading to operational inefficiencies and increased surface handling costs. Therefore, proactive intervention is essential to preserve oil productivity and extend well life.

One effective strategy for mitigating high GOR is the application of Selective In-Situ Gas Shut-Off Technology (SIGSHOT) treatments. These treatments aim to selectively block gas flow from specific intervals without impairing oil production. Success depends on accurate identification of gas entry points, typically achieved through high-resolution production logging using tools such as spinner flowmeters and optical sensors. Once identified, chemical agents—such as polymer gels or relative permeability modifiers —are injected into the reservoir. Under in-situ conditions, these agents form barriers that restrict gas movement while allowing oil to continue flowing.

In addition to chemical interventions, completion optimization plays a vital role in managing gas production. Techniques such as, intelligent well completions enable operators to control inflow from different reservoir layers. Avoiding well placement near the GOC and focusing on oil-rich zones can significantly reduce the risk of gas breakthrough. Moreover, the use of downhole flow control devices and real-time monitoring systems allows for dynamic production management based on changing reservoir conditions.

Advanced reservoir simulation models further support decision-making by predicting gas cusping behavior and evaluating the impact of various production strategies. These models integrate geological, petrophysical, and fluid flow data to simulate reservoir dynamics and guide optimal well design and intervention planning.

In summary, while pressure maintenance through gas and water injection is essential for reservoir performance, wells near the GOC require targeted strategies to manage gas production. Through a combination of diagnostics, chemical treatments, optimized completions, and simulation, operators can effectively mitigate gas cusping and enhance oil recovery.

## Methodology

For high-GOR wells, optimal drawdown is determined through multi-rate production testing. A more effective long-term solution involves isolating the free gas-producing zones—typically through a Gas Shut-Off (GSO)—allowing for higher drawdown to be applied to the oil-producing zones without exacerbating GOR issues.

Therefore, it is urgent to perform remedial treatments to control excessive gas production and enhance oil recovery. Implementing gas shut-off in high-GOR wells enables increased oil production within the same gas handling capacity. However, the high-temperature and high-salinity conditions of the reservoir pose significant challenges in selecting suitable gas shut-off chemicals. Additionally, conventional gas shut-off technologies involving complex wireline interventions are both costly and time-consuming.

To address these challenges, an alternative solution was sought —one that minimizes operational time and cost. As a result, a rig less gas shut-off method, known as Selective In-Situ Gas Shut-Off Technology (SIGSHOT), is proposed for pilot testing in the Field."

It is also important to note that excessive gas production can lead to additional operational challenges, such as increased erosion rates in production flowlines and limitations imposed by topside facility gas handling constraints:

The following is a summary of the surveillance data that must be reviewed prior to planning a gas shut-off operation.

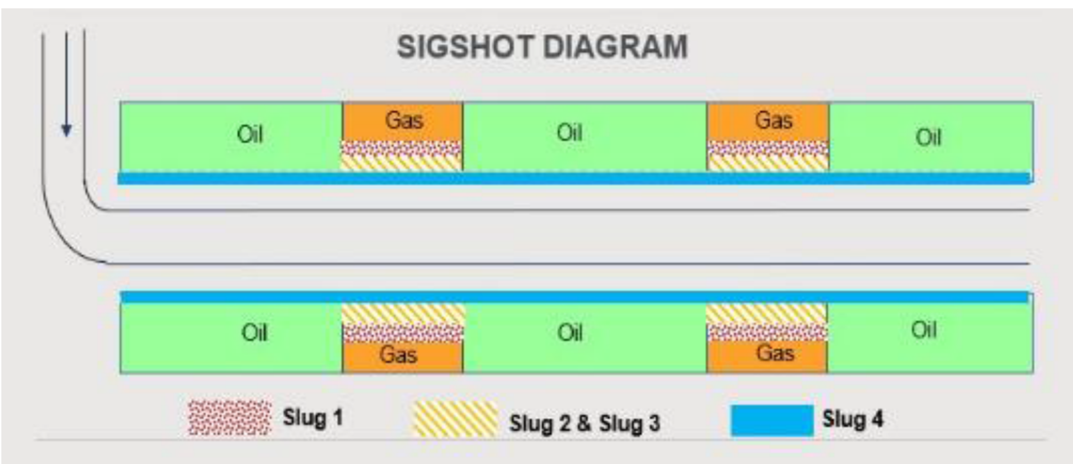
1. Identifying the Depth of Free Gas Production Zone: Determining the depth of the free gas production zone is crucial. This can be achieved through a production logging survey on the well, complemented by existing log data from nearby offset wells, if available. Ideally, well equipped with fiber optic sensors across the field can provide continuous monitoring, eliminating the need for production logging interventions to pinpoint the gas source.
2. Well Drawdown Considerations: Wells operating under low drawdown conditions tend to have a higher probability of success and extended effectiveness following a gas shut-off intervention.
3. SIGSHOT (Gas Shut-Off) Concept and Economic Evaluation: The SIGSHOT technique involves isolating previously producing intervals that have been compromised by gas cusping. It is essential to assess the impact of gas shut-off on overall reserves recovery as part of the economic justification for the well intervention. This evaluation should also consider the potential to produce the isolated reserves through alternative wellbores.

## Technology Description

Selective In-Situ Gas Shutoff Technology (SIGSHOT) is a rigless gas shut-off (GSO) method designed to isolate gas-producing zones without the need for a rig-based intervention. This chemical shut-off technology requires a series of carefully planned steps to address the high gas/oil ratio (GOR) in the targeted wells. The primary goal was to implement a Selective In-Situ Gas Shut-Off Technology (SIGSHOT) treatment to mitigate gas influx and enhance oil recovery.

1. **Baseline Production Logging:** The operational workflow began with a baseline production logging campaign. This step was essential for identifying zones with excessive gas contribution. Advanced logging tools, including spinner flowmeters and optical sensors, were used to accurately characterize the flow profile and isolate high-GOR intervals.
2. **Selection of Chemical Agents:** Following the diagnostic phase, a chemical treatment was applied to the identified zones. The selection of chemical agents was based on their compatibility with reservoir conditions and their ability to selectively block gas flow without impairing oil production.

3. **Implementation of SIGSHOT:** The SIGSHOT technique involves isolating previously producing intervals that have been compromised by gas cusping. The treatment involves the sequential injection of four main slugs:
   - **Slug 1 - Preflush and Gas-Controlled Self-Blocking Agent:** Prepares the formation by conditioning the near-wellbore environment and initiating partial blockage of gas flow paths through a self-regulating mechanism.
     - Treatment Depth ~10 M
   - **Slug 2 - Primary Gas Shut-Off Agent:** Forms an in-situ gel within the near-wellbore region, creating a robust barrier to prevent gas channeling.
     - Treatment depth ~3-5 m
   - **Slug 3 - Reinforced Gas Shut-Off Agent:** A higher-concentration in-situ gel designed to strengthen the gas blockage, particularly in zones with elevated pressure gradients near the wellbore.
     - Treatment depth ~3-5 m
   - **Slug 4 - Over-Displacement Agent (Water):** Facilitates deeper penetration of the injected chemicals into the formation, ensuring broader coverage and improved effectiveness of the gas shut-off treatment.




## Well Information

The well completed in crestal location in the field, with production started from march, 2021, before the implementation of SIGSHOT treatment, the well experienced sharp increase in GOR surpassing the reservoir management Guidelines and resulting in shut-in the well (Fig.1).

Production logging serves as a vital diagnostic tool for evaluating the distribution of fluids along the wellbore and pinpointing zones for targeted interventions. When chemical shutoff treatments are frequently considered. To make these interventions as effective as possible, a thorough understanding of the well's behavior before and after treatment is crucial.

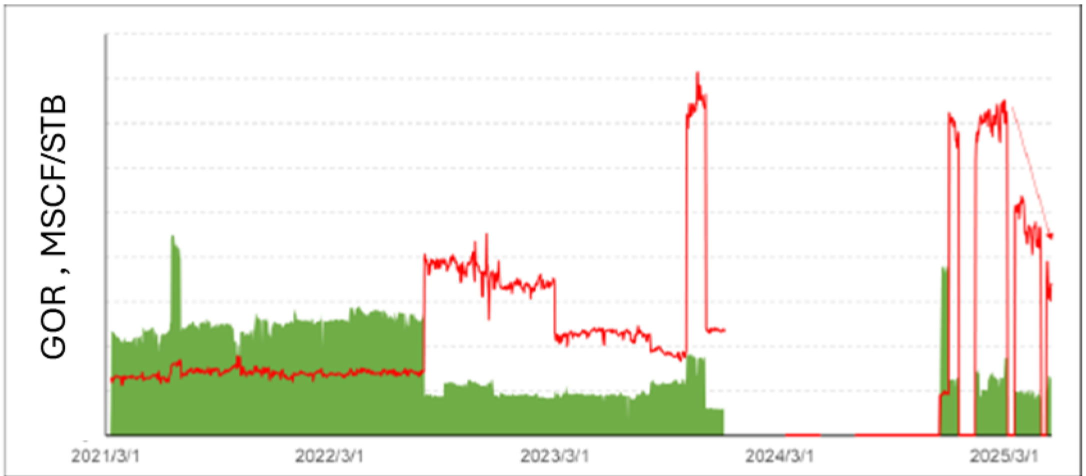



**Figure 1—illustrates the production history of a typical well before and after the onset of increased GOR. As gas production rises, the well's productivity index is negatively impacted, as free gas occupies portions of the completion interval, restricting liquid inflow. This also affects the well's target drawdown. In most cases, drawdown must be limited to prevent further GOR escalation and the resulting accelerated oil decline.**

## Pre-Treatment Production Logging

Production logging was performed before any remedial action to establish a baseline understanding of the well's inflow profile, as well as to identify problematic intervals. The main objective is to identify which intervals in the wellbore contribute the most to gas production. By deploying production logging tools such as spinner flowmeters, temperature and optical sensors, it becomes possible to accurately pinpoint the depths responsible for the most significant gas influx. This information is critical for ensuring that any chemical treatment is precisely targeted. Furthermore, production logging data allows for the quantification of the relative contributions of gas, oil, and water from various reservoir sections. Such data is essential not only for guiding the placement of shutoff chemicals but also for predicting post-treatment performance. Insights gained from this data also helped characterize reservoir heterogeneity, such as differences in permeability, flow barriers, or thief zones, all of which refine the treatment strategy—especially in complex reservoirs where gas breakthrough is exacerbated by such heterogeneity. Ultimately, these activities established a clear baseline production profile, which became a vital reference point for measuring the success of the chemical shutoff treatment later on.

The high gas-producing intervals were identified through production logging were zone A, B, C and D, the chemical shutoff treatment was tailored to the well's specific needs (Fig.2 and Table 1). The selection of chemicals was based on the reservoir characteristics and the nature of gas pathways, as identified through PLT data, which revealed the severity and depth of gas entry points. Accurate identification of these intervals ensured that chemicals are injected exactly where needed, which minimized wastage and maximized the effectiveness of the treatment. Additionally, the production profile assisted in determining the optimal volumes and concentrations of chemicals required to achieve an effective gas shutoff, all while avoiding any impairment of oil production from desirable zones.

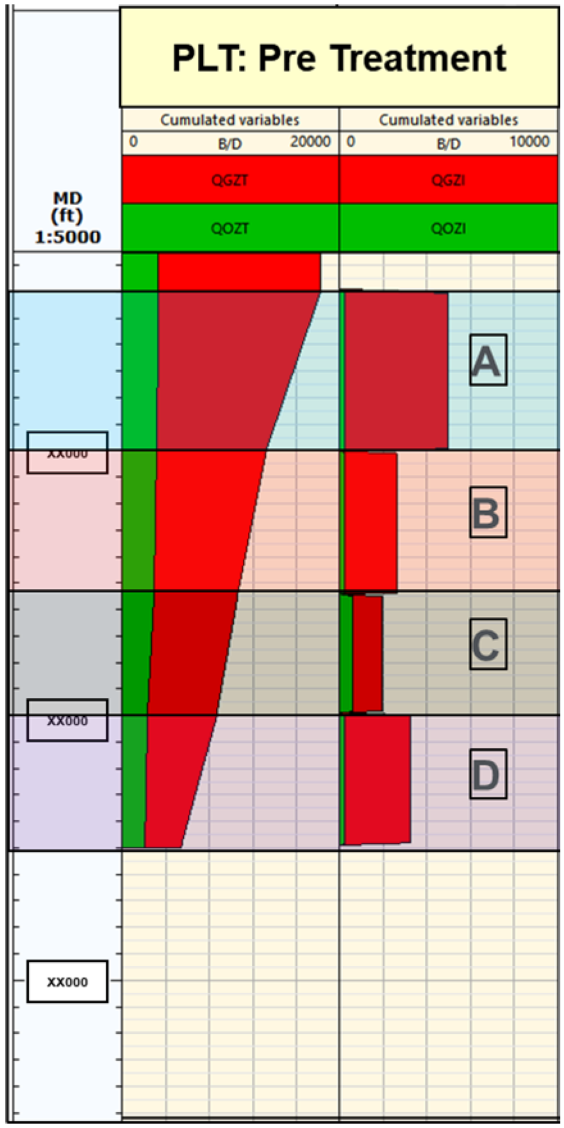



Figure 2—Pre treatment production logging data illustrates zones A, B, C and D are high gas producing intervals.

Table 1—PLT results prior to the treatment

| Zone | Oil% | Gas% |
|---|---|---|
| A | 5.9 | 32.2 |
| B | 6.9 | 16 |
| C | 18.2 | 8.9 |
| D | 6.9 | 20.4 |

In JUN 2024, the Chemical Gas shut-off were performed and the operation was basically successful with no safety incidents Fig.3)

Before treatment, before treatment the well experienced decline in oil rate and harsh increase till eventually closed due to exceeding the reservoir management GOR guideline and resulted in well shut-in.

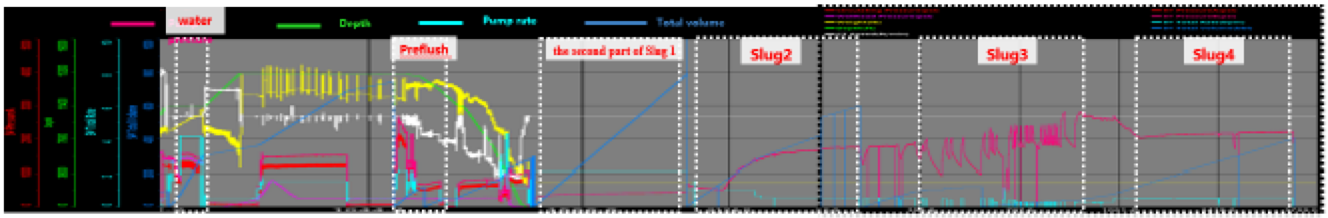



**Figure 3—Dynamic parameters in different stages of SIGSHOT treatment**

## Post-Treatment Production Logging

After the chemical shutoff is completed, a second round of production logging was carried out to evaluate the outcome of the intervention. The primary goal at this stage is to assess whether the previously targeted intervals now show reduced or eliminated gas flow compared to the baseline profile. Equally important is confirming that the chemicals have not inadvertently migrated or impaired flow in oil-productive intervals that were not intended for treatment. Post-treatment production logging can also reveal whether any new production anomalies have emerged, such as newly developed zones of gas or water influx due to changed reservoir dynamics. The comparative analysis of production logging data before and after the chemical shutoff is crucial for evaluating the success of the intervention. Overlaying the flow profiles from before and after the treatment allowed for a direct assessment of the impact. It is evident from the comparative analysis of pre & post production logging data illustrated a nil or negligible gas flow rates from previously high gas-producing intervals (A, B and C), with oil production in desirable zones maintained or even improved. Additionally, the inflow profile demonstrated that untreated zones remain unaffected, confirming the selectivity of the treatment. All of these downhole observations corroborate by surface production rates and gas-oil ratios.

PLT results show that the main gas production interval has been blocked (Fig.4 and Table 2 and 3)

**Table 2—PLT results after the treatment**

| Zone | Oil% | Gas% |
|---|---|---|
| A, B &C | 0 | 0 |
| D | 26.5 | 82.5 |
| E | 73.5 | 17.5 |

**Table 3—Production test before and after SIGSHOT**

| Test Date Time | Test Type | Choke size | Net Oil Rate % of well Potential | GOR % Of RM GL | Remark |
|---|---|---|---|---|---|
| 18-Oct-23 | PTS | 48 | 79 | 133 | |
| 20-Oct-23 | PTS | 68 | 100 | 124 | Before SIGSHOT |
| 23-Nov-23 | PTS | 32 | 51 | 134 | |
| 14-Dec-24 | PTS | 32 | 61 | 105 | |
| 5-Dec-24 | PTS | 34 | 73 | 113 | |
| 15-Dec-24 | PTS | 42 | 76 | 118 | |
| 26-Feb-25 | PTS | 16 | 25 | 50 | After SIGSHOT |
| 27-Feb-25 | PTS | 24 | 56 | 82 | |
| 28-Feb-25 | PTS | 32 | 79 | 96 | |
| 1-Mar-25 | PTS | 42 | 96 | 114 | |

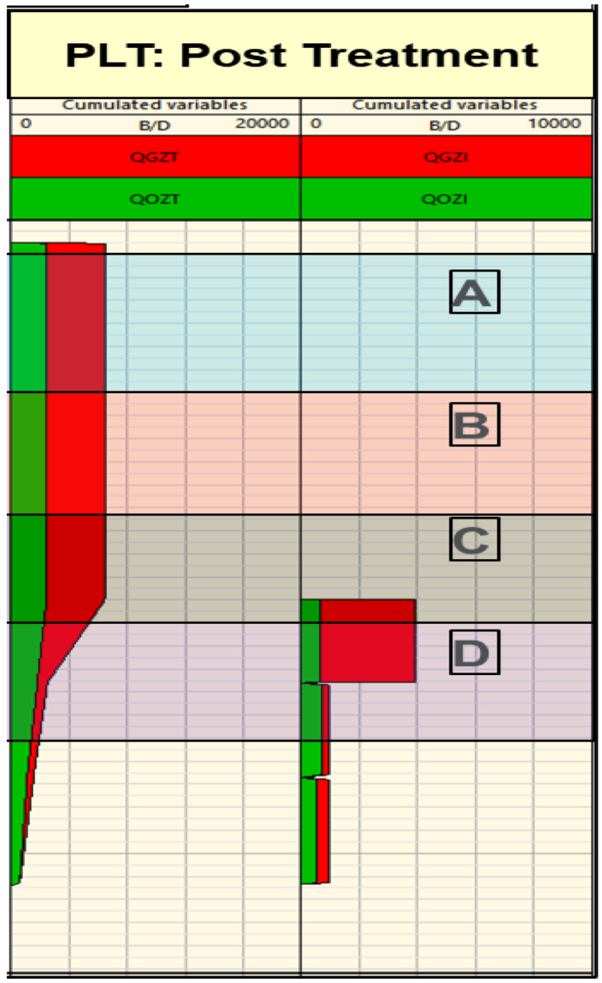



**Figure 4—Post treatment production logging data illustrates zones A, B, C and D are high gas producing intervals.**

Table 3 shows the comparison of GOR and oil production rate Pre and Post SIGSHOT treatment from PTS results

According to the current data, based on valid PTS results conducted at choke 32, GOR has decreased by over 25%, indicating that the SIGSHOT technology has been successfully implemented.

## Results

The pilot implementation of the Selective In-Situ Gas Shut-Off Technology (SIGSHOT) in Well X demonstrated clear operational success. Prior to treatment, the well exhibited high GOR., exceeding operational guidelines and necessitating shut in. Production logging identified four key intervals (Zones A-D) contributing to excessive gas influx.

Following the chemical intervention:

- Post-treatment logging confirmed a substantial reduction in gas flow from the targeted zones.
- Oil production rates improved or remained stable, with no adverse impact on untreated intervals.
- GOR decreased by over 25%, stabilized conformance and preserved reservoir energy.
- The well was successfully reactivated, sustaining an oil production rate of approximately 80% of well the production potential and successfully operated with the reservoir management guidelines.

These results validate the precision and effectiveness of the SIGSHOT approach in isolating gas-producing zones while preserving oil productivity.

## Conclusions

The SIGSHOT pilot has proven to be a reliable, rigless solution for managing high GOR in carbonate reservoirs under WAG injections. By leveraging high-resolution diagnostics and targeted chemical treatments, the intervention successfully mitigated gas interference, enhanced oil recovery, and extended well life.

The importance of this chemical gas shutoff technology cannot be overstated. It addresses one of the most challenging aspects of reservoir management—controlling high gas/oil ratios (GOR) that can significantly impair oil recovery and operational efficiency. The ability to selectively block gas flow from specific intervals without impairing oil production is a game-changer for the industry. This technology offers a scalable solution for similar reservoir challenges and supports broader objectives for efficient and sustainable field development.

The effectiveness of the SIGSHOT approach is evident from the results of the pilot study. The targeted chemical treatments significantly reduced unwanted gas production, as confirmed by post-treatment production logging data. The reduction in gas contribution from the treated zones led to stabilized or improved oil production rates, demonstrating the success of the intervention. Additionally, the rigless deployment of this technology minimizes operational costs and complexity, making it a cost-effective solution for managing high GOR in WAG-injected carbonate reservoirs.

Post-treatment validation confirmed the selectivity and effectiveness of the chemical in-situ gas shutoff technology. The comparative analysis of production logging data before and after the treatment showed nil or negligible gas flow rates from previously high gas- producing intervals, with oil production in desirable zones maintained or even improved. These downhole observations were corroborated by surface production rates and gas-oil ratios, further validating the success of the intervention.

In summary, the SIGSHOT technology offers a viable and effective solution for managing high GOR in carbonate reservoirs. Its ability to selectively block gas flow, combined with the cost-effectiveness of rigless deployment, makes it an attractive option for operators facing similar challenges. The success of this pilot project demonstrates the potential of this technology to enhance oil recovery, stabilize production rates, and extend the productive life of wells, ultimately contributing to more efficient and sustainable reservoir management.

This approach offers a scalable solution for similar reservoir challenges and supports reservoir broader objectives for efficient and sustainable field development.